

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n, i;

    printf("請輸入一個整數: ");
    scanf("%d", &n);

    printf("%d 的因數有: ", n);

    for (i = 1; i <= n; i++) {
        if (n % i == 0) {
            printf("%d ", i);
        }
    }

    return 0;
}
```

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```
iseng@BadApple MINGW64 /f/Yuntech/p/class/0407HW (main)
$ ./HW7-2_6.exe
請輸入一個整數: 24
24 的因數有: 1 2 3 4 6 8 12 24
iseng@BadApple MINGW64 /f/Yuntech/p/class/0407HW (main)
$ |
```

```
programming class > 0407HW > C HW7-2_10.c > main()
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int main() {
5      int i, n;
6      float sum=0;
7
8      printf("請輸入n值:");
9      scanf("%d", &n);
10     for (i=1; i<=n; i++) {
11         sum=sum+1.0/(float)i;
12     }
13     printf("總和為:%f", sum);
14 }
```

MINGW64:/f/Yuntech/pro

```
iseng@BadApple MINGW64 /f/Yuntech/pro
$ ./HW7-2_10.exe
請輸入n值:3
總和為:1.833333
iseng@BadApple MINGW64 /f/Yuntech/pro
$ |
```

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int i, n, sum;

    for (n=2; n<=1000; n++){
        sum=0;
        for (i=1; i<n; i++) {
            if (n%i==0) {
                sum+=i;
            }
        }
        if (sum==n) {
            printf("%d是完美數\n", n);
        }
    }
}
```

MINGW64:/f/Yuntech

```
iseng@BadApple MINGW64 /f/Yuntech
$ ./HW7-2_11.exe
6是完美數
28是完美數
496是完美數
iseng@BadApple MINGW64 /f/Yuntech
$ |
```

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n, a, b, c, sum;

    for (n=100; n<=999; n++) {
        a=n/100;
        b=(n/10)%10;
        c=n%10;

        sum= a*a*a+b*b*b+c*c*c;

        if (sum==n) {
            printf("%d是阿姆斯壯數\n", n);
        }
    }
}
```

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iseng@BadApple MINGW64
\$./HW7-2_12.exe
153是阿姆斯壯數
370是阿姆斯壯數
371是阿姆斯壯數
407是阿姆斯壯數

iseng@BadApple MINGW64
\$ |

programming class > 0407HW > C HW7-2_32.c > main()

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int main() {
5      int n, i, P=1;
6
7      printf("請輸入一個數字:");
8      scanf("%d", &n);
9      for (i=2; i<n; i++) {
10         if (n%i==0) {
11             P=0;
12             break;
13         }
14     }
15     if (P) {
16         printf("此數是質數\n");
17     }
18     else {
19         printf("此數不是質數\n");
20     }
21 }
```

MINGW64:/f/Yuntech/progr

iseng@BadApple MINGW64
\$./HW7-2_32.exe
請輸入一個數字:32
此數不是質數

iseng@BadApple MINGW64
\$./HW7-2_32.exe
請輸入一個數字:17
此數是質數

iseng@BadApple MINGW64
\$ |